

From Ruin to Rehab



An architect and a boiler mechanic team up to reconstruct tight, well-ventilated, healthy homes from 18 abandoned buildings.

by Henry Gifford

Last spring, the first energy-efficient residential buildings went on sale in New York City. Just three years earlier, these homes were badly deteriorated wood frame and masonry structures that had been abandoned by the owners because they had become uneconomical under New York City's rent control law. The buildings were in such poor shape at the start of the project that they required major, or "gut," rehabilitation,

and the work done to them blurred the distinction between renovation and new construction. In most of these buildings, the exterior walls were the only part that was kept.

I worked with architect Chris Benedict as she designed and supervised the rehabilitation of these abandoned houses into multifamily homes of two, three, or four apartments each. Each building was sold to people who were expected to live in one apartment and

rent out the others. The reasons that the existing buildings were used instead of being knocked down were that their floor areas sometimes exceeded what would be allowed under current zoning rules, and that the buildings' owner wanted to maintain the fabric of the neighborhood.

The developer for the project was NY/Enterprise CityHome HDfC. The buildings were given to the developer by New York City's Department of

Housing Preservation and Development. The developer asked Benedict to meet two, fairly typical design criteria:

1. Get the job done as cheaply as possible.
2. Meet code.

Benedict decided to make this batch of CityHomes as sustainable as possible. At first, the Department of Housing Preservation and Development objected. They were concerned that making the project sustainable would add cost, disrupt the bid process, and make the homes too difficult for the contractors to construct and for the homeowners to operate.

After much discussion, Benedict, with the help of Andy Padian of the Association for Energy Affordability, got all parties to agree to the following bid method:

- The contractors would bid on the code minimum job.
- The lowest-bidding contractor would be required to submit a detailed trade payment breakdown of the bid. Benedict would be allowed to modify the typical bid sheet in order to get the contractor's prices for items that she would ultimately exchange for more sustainable materials and technologies.
- Once the contractor's bid was deemed satisfactory, Benedict would introduce a package of sustainable alternatives broken down into the price categories provided by the contractor in the trade payment breakdown.

John Frezza of Strategic Construction in Brooklyn submitted the winning bid of \$4,100,000. He asked for \$54,000 on top of the original bid for the trade-offs that Benedict wanted to substitute (see Table 1). While this is only about 1% extra, it was not the zero extra cost that Benedict had hoped for. Eventually, NY/Enterprise City-Home HDFC received a \$37,000 grant from the Joyce Mertz Gillmore Foundation and a matching grant from

NYC Housing Preservation and Development that together more than covered this extra cost.

Designing the Building Envelope

Since the biggest impact a building has on the environment over the long run is its energy use, Benedict decided to focus on making the buildings more energy-efficient, but she also specified using recycled or less toxic materials whenever possible. Some of the less toxic products she specified included exterior grade plywood, masonite faced interior doors, and latex caulk. She designed the houses using the systems approach, with a strong emphasis on the interaction of three main systems: the building envelope, the ventilation system, and the heating and cooling system.

Benedict designed the buildings so that there would be an air barrier separating each apartment from any area outside the apartment, and insulation between each apartment and any unheated areas. The air sealing was done using the airtight drywall approach, which Benedict first learned about in an Affordable Comfort conference presentation given by Maureen Davlin and Paul Knight of Chicago (see

"Chicago Apartments Get New Lease on Life," *HE* Mar/Apr '97, p. 23).

In these houses, the air barrier was made continuous by connecting the drywall to the floors with a bead of caulk. The normal taping and joint compound make the screw holes and seams sufficiently airtight and, at the top, the drywall was extended up to the



After first cutting the drywall to fit around the joists, the contractor figured out that stopping the drywall below the joists and then adding scraps that fit each opening was a more efficient method of constructing a continuous air barrier.

subfloor above and caulked. The caulking and the extension of the drywall above the ceiling were the only costs added to achieve the continuous air barrier. However, since drywall extended up to the subfloor is one of two fire-stopping methods approved for use in New York City, these costs were partly offset by avoiding the need for any other fire-stop.

Table 1. Cost Tradeoffs Associated with the Design of These Houses

First-Cost Increase (Over Code Minimum)	First-Cost Decrease
Air sealing	Smaller heaters in each room (30%–50% saving)
Slightly more than normal insulation and better than normal insulation installation (\$1.30 per ft ²)	Smaller boiler, pump, pipes, etc. (small saving)
Low-e windows (\$20 more per window)	No boiler room required (\$2,000 saving)
	No fresh-air intake required (\$200 saving)
	No chimney to the roof required (\$1,500 saving)
Active ventilation system (no net cost increase)	Replaced bathroom/kitchen fans
	Simpler wiring, no timer needed
	Duct fireproofing not required
Thermostats in each room (\$40 more per thermostat)	
More planning	Fewer callback expenses for the builder
More supervision	Smaller, cheaper mechanical systems made possible by avoiding oversizing resulting from vague size estimates

Sealing the Penetrations

To ensure the completeness of the air barrier, every place a pipe or wire passed through the drywall or subfloor, the hole was carefully caulked or foamed. The electric boxes were another question. Benedict looked at samples of airtight electric boxes or liners and worried about the holes that would be made for the wires. She rejected airtight cover plates as being too easy for rental tenants to remove. We settled on using regular electric boxes and smearing them with a layer of duct mastic after the wires were installed but before the drywall went up. This worked out well; relatively unskilled workers using a piece of cardboard as a flexible spatula made the boxes airtight at little cost. Later, when the drywall was installed, a bead of caulk was applied to the front of the boxes.

For insulation, blown-in cellulose was dense-packed into a closed cavity. Benedict decided not to use a vapor barrier after an analysis of the temperature gradient through the walls using the MOIST computer program indicated that there would be negligible condensation in the walls. With no vapor barrier, the walls can dry to either side if moisture or liquid water does get in.

After the air tightening was completed and the insulation had been installed, Benedict used a blower door to measure air change rates in each unit. The ACH at 50 Pascals (Pa) ranged from 2.3 to 6.7, with an average ACH50 of 4.4.

Designing the Ventilation System

For ventilation, Benedict chose to use constantly running individual exhaust-

only systems for each apartment. Normally, buildings of this type in New York City would have a common exhaust duct for all of the apartments, with connections at each bathroom leading to a common fan on the roof. Such a system poses numerous problems. It doesn't always succeed in pulling air from each apartment; it is next to impossible to balance; and it might be turned off soon after the building is inspected. In winter, the stack effect can cause exhaust air from lower floor apartments to flow into the ductwork and out into upper floor apartments. Finally, the common exhaust ducts allow annoying sound crosstalk between apartments.

To supply fresh air, Benedict chose to have trickle ventilators installed in each bedroom, in the windows located directly above the heaters. A trickle ventilator is a slot in the window that is fitted with an insect screen, a hood to prevent rain from entering, and a movable plastic blade that limits air to a fixed flow regardless of pressure differences caused by wind or other factors.

Each apartment has one exhaust fan that pulls air from ducts connected to the kitchen and each bathroom in that apartment. The fans are fed power from the circuit breaker next to the refrigerator circuit breaker, and the two circuit breakers are tied together to discourage occupants from turning the fans off. Constant air flow regulators were installed at each connection to the ductwork at the kitchens and baths to set the air flow to what Benedict calculated would meet both code requirements and actual health needs.

Although we strongly wanted to include range hoods in the kitchen, they were out of the question, both for financial reasons and because we doubted that occupants would use them appropriately. The kitchen exhaust port runs negative to the apartment, because it has the highest CFM

draw, affording the kitchen 50 CFM of continuous ventilation. While not ideal, this level of kitchen ventilation is acceptable.

We chose to use round metal ductwork because it is inexpensive, fire-resistant, and easy to seal, and because the friction between it and the moving air is low and predictable. I calculated the sizes of the ductwork according to air-



The exhaust ventilation system pulls outdoor air through the trickle ventilators in each bedroom, through the common areas, and out each bathroom and kitchen.

flow requirements using ACCA's *Manual D*. The size and location of every piece of equipment, including the ductwork, was shown on the plans. Benedict specified that the ductwork would be sealed with mastic.

Benedict wanted to run the ventilation exhaust out the sidewall, but New York City code requires bathroom ventilation to be run to the roof unless there is a window, in which case no mechanical ventilation is required. Benedict obtained a waiver from the Buildings Department by arguing that a bathroom with a window doesn't vent to the roof, so why should these! Only the top-floor apartments have roof fans. The middle floors have inline fans accessed by panels in the ceiling, and lower-floor apartments have fans mounted outside if they are low enough that the fans can be repaired without needing an exceptionally long ladder. However, both the roof fans and the outdoor fans have turned out to be too cumbersome and noisy, even after \$30 mufflers were installed in all systems. Next time we will specify the Panasonic inline fans only, as they are much quieter (see "Next Time").

The system as installed pulls outdoor air through the trickle ventilators in each bedroom, through the common

Next Time

Based on lessons we learned in this project, in the future we would specify:

- Panasonic inline fans only;
- an insulated domestic hot water return line with a Taco 003 pump running nonstop; and
- air tightening the circuit breaker box with duct mastic.

If there had been more money available, we would have specified:

- ductless air conditioning; and
- dehumidification.

areas, and out each bathroom and kitchen. An exhaust-only ventilation system can create indoor air quality (IAQ) problems under certain conditions (see "Forum Brings Fresh Air to Ventilation Problems," p. 15). However none of those problematic conditions existed in these houses. These houses have sealed-combustion appliances and no attached garages, and radon is not a problem for most areas of Brooklyn.

Whatever the ventilation system's imperfections may be, it is expected to perform better than anything else within the budget. Any extra costs are partly offset by the avoided cost of the bathroom ventilation system required by code for bathrooms without windows.

Designing the Heating System

To get a well-functioning heating system, we used standard equipment, but noted each piece of equipment and pipe and its part number, size, and location on the plans. We also included a wiring diagram and a list of settings for every control. Benedict did a heat load calculation for each room using ACCA's *Manual J*. The most unpredictable part of any heat load calculation is guesstimating the infiltration, or air leakage through the building. The air tightening reduced the heating capacity needed to meet the infiltration load in two ways: It reduced the actual load and it made the load; more predictable, reducing the need for higher safety factor margins.

Benedict's calculations showed that direct-vent, sealed-combustion appliances could meet the reduced heat loads. A typical heating load for comparably sized buildings would have been 30%–50% higher, but the boiler and baseboards would have to be substantially larger, since heating systems are not usually designed for the actual loads. Larger heating loads would have meant using conventional chimney vented appliances, which cost a few hundred dollars less but require a new chimney to the roof that costs thousands of dollars. New York City code requires a boiler room in buildings housing three families or more. By using sealed combustion boilers, Benedict was able to get a waiver of this requirement. The boilers ranged in size from 90,000 Btu/hr input to 175,000 Btu/hr input for the largest four-family houses.

We chose to specify hot-water heat to avoid the discomfort, health problems, and noise associated with scorched-air heat. Also, with hot-water heat, it's easy to control the temperature in each room individually—a necessary requirement for an energy-efficient building. We chose baseboard heaters because the first cost is much lower than for any other option. Thermosta-



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tic radiator valves, which reduce or stop the flow of water as the room temperature increases, were specified for every room. (These valves should be used only on systems where the piping is designed for proper flow.) The owners can control the temperature by turning

a knob, but the valves in the rental apartments are locked at about 74°F.

The plans showed the sizes of all the heaters, their locations (where they were least likely to be blocked by furniture), the locations and sizes of heating pipes, the locations of the boilers and their vents, wiring diagrams for the heating control system, piping diagrams for the near-boiler piping, and settings for all the controls.

We chose to use a Heat-Timer HWM-100 control, which changes the temperature of the heating system water in response to the outdoor temperature. This control cost almost \$300—a cost that was partly offset by not having to install relays for the two pumps in the system. An optional clock reduces water temperature at night to provide night setback. We chose this control because the outdoor temperature at which the control automatically turns the heating system on and off can be adjusted separately from the amount of heat delivered to the house, an option that many other controls of this type do not provide. An inexpensive Danfoss Thermic Valve installed near the boiler protects it from condensation that otherwise could occur when the system runs at low temperatures for a long time.

We chose Weil-McLain GV boilers for most of the houses, because they are generally very safe and reliable. Although Weil-McLain's fan motors need regular lubrication, the system requires no other regular maintenance. I tested many of the boilers and consistently found 8%–8.5% CO₂ and about 15 ppm CO in the flue gases. These are very safe levels.

I sized the heating boilers by adding up all the room loads, not by adding up the capacities of the installed radiators.

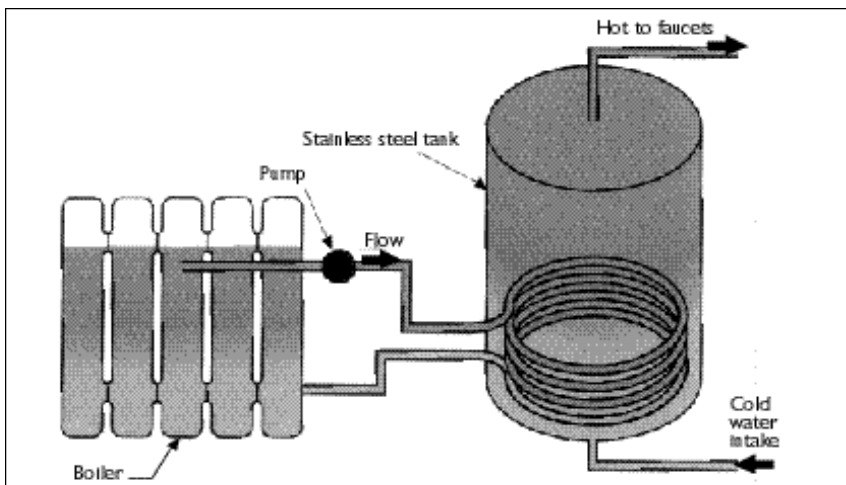


Figure 1: Hot water from the boiler is pumped through the indirect water heater in the domestic hot-water storage tank. (See photo on p. 28.)

If, for example, the heat load calculation showed that a room needed 6.2 ft of baseboard heater, I had a choice between specifying 6 ft or 7 ft. Six wasn't enough, so I specified 7, as I knew the thermostat could prevent overheating. But I used the heating capacity of 6.2 ft of heater to size the boiler, not the capacity of the 7 ft installed. In one case where I had to choose between a

boiler that was slightly smaller than the heat load and the next size, which was significantly larger, I chose the smaller one, because it's impossible for all the rooms to have peak heating loads simultaneously. Peak heat load calculations assume that cold outdoor air is leaking into the room, but in fact air cannot be entering all rooms simultaneously; if it enters some rooms, it must also be leaving from other rooms.

Hot Water at the Ready

No boiler capacity was added to accommodate domestic hot water load, for three reasons. First, each house had significant storage of hot water; second, heat loads peak at around 5 am, when few people shower; and third, the heating system control automatically turns off the heating system pump for the few minutes it takes to warm up the hot water. With this "hot-water priority" strategy, water is reheated several times faster than it is by a normal direct-fired water heater (see Figure 1).

I selected Stor-ex indirect water heaters, because they are made of stainless steel throughout. They are guaranteed for 15 years and are basically limeproof because any lime in the water will collect on the outside of the coil, where the thermal shock created when the pump comes on will cause it to break off. These heaters ranged in size from 60 gallons for the smallest two-family home to 115 gallons for the



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largest building.

Thermal layering, or stratification, of water in the tank can be a problem, but it is easily prevented by installing a well insulated return line from the top of the hot-water riser back down to the cold-water inlet to the water heater, and installing a small pump to circulate the water slowly. A small pump like a Taco 003, wired to run continuously (47 watts), moves the water slowly enough to allow people to shower with the hot water in the top of the tank even as cold water is being heated in the bottom of the tank, yet it still moves the water fast enough to prevent stratification. The energy used by the pump and the return piping is less than what would be lost by using hotter water and from running leaks caused by faucet washers damaged by overheated water. Return lines weren't specified in these houses, because we expected them to be installed as a standard item. After the houses were occupied, a hot water stratification problem alerted us to the fact that return lines had not been installed. In the future, we will specify them.

The combination of an indirect water heater and a direct-vent, sealed-combustion boiler wired for outdoor reset allows for maximum efficiency and reliability. (This combination is standard for a new heating system in Europe.) The heat losses from the indirect water heater through the insulation are small, and the heat is not lost in the winter since it warms the basement.

Since the indirect water heater has no chimney or vent of any type, the largest path for losses in a normal water heater is eliminated. In the wiring diagram on the blueprints, the boiler is wired for cold start, which means that it stays cold unless there is a call for heat or hot water. For example, all summer it stays cold except when it is actually heating the indirect water heater. The time it takes to heat up is so short that there is no significant delay on a call for heat or hot water.

Heating-system pipes in basements were left uninsulated to give the basements a little heat. Heating pipes in the rest of the house were installed inside the thermal envelope and left uninsulated. All the hot-water pipes were insulated to save energy in the summer, and cold-water pipes were insulated to prevent condensation.

Construction

To ensure that the buildings were actually constructed as designed required more supervision than the average job receives. For example, Benedict not only specified window make and model but also specified in detail how the windows were to be installed. The contractors were told ahead of time that their work would be inspected.

Careful attention to details that could not be fixed later—such as air sealing—paid off. The contractors' workers who did the walls, floors, and air sealing quickly learned the air-tightening methods, surprising everyone with how easily that part of the job went.

The mechanical subcontractors were another story. They took the plans to be a vague suggestion, like lane markers on New York streets, instead of a description of what they had contracted to do. They had agreed to install one heating system for examination before installing any more, but instead they quickly installed as many as possible with little regard for the specifications, knowing that it is rare on most jobs for anything to get changed once it has been installed. One or two refusals to sign payment requisitions cured them of this, but it was a painful process.

Testing and Inspection

Benedict used a blower door to test the air tightening in each apartment. All but two apartments passed the first time, and those two were easy to fix. One apartment had a large hole where the gas clothing dryer vent pipe went outdoors, and the other had a hole around the wire to the kitchen exhaust register. The main remaining path for leakage into each apartment turned out to be the circuit breaker box. Sealing the breaker box with duct mastic is on the “next time” list.

We used an infrared camera to scan the walls to see if the insulation was installed properly. The insulation contractors returned and fixed the insulation where the scanning showed that the contractors hadn’t filled the cavities completely. Benedict tested the ventilation system by building a pressure pan out of a cardboard box and some foam rubber. All the ventilation system inlets were found to be running very close to the specified flow rates when the correct constant air flow regulators were installed.

The boilers were checked to make sure they were running at the specified firing rate by checking the actual size of the flame. This was done simply by timing how many seconds one revolution of the test dial on the gas meter, which is labeled for the ft^3 of gas per turn, took. Since 1 ft^3 of gas is almost exactly 1,000 BTU, it’s easy to calculate the hourly firing rate and compare it to the input firing rate data on the boiler’s label. Some of the boilers needed adjustment because New York City has lower gas pressure than most of the country, and equipment is set up for the normal pressure range when it is sold. No doubt the unpredictability of the actual firing rate without this check has reduced people’s faith in heat load calculations over the years—which is unfortunate, since anyone with a calculator and a watch can do this test in a few minutes.

We checked the air pressure in the expansion tank, because if the contractors do not provide the extra pressure necessary to accommodate the height of the buildings, air in the top-floor radiators can create problems in the heating system. We checked all the heating-system control settings to



Chris Benedict measures the ventilation air flow with a manometer and a homemade pressure pan.

ensure that they were set correctly and will require absolutely no adjustments.

How Well Did the Houses Work?

In the “test by fire” category, someone smelled smoke on a Friday in one of the houses, but investigators found no fire. The following Monday, someone found a fire smoldering in the ceiling. The firefighters tore down the ceiling and found the cellulose insulation smoldering. They hosed the place down with a generous amount of water. Since the fire consumed only a few ft^2 of insulation over a weekend, the air tightening obviously made the houses more fireproof than they would have been otherwise. The cause of the fire was never determined.

The water from the hoses thoroughly soaked the walls and insulation in that room, providing an opportunity to see how the walls handled moisture. We were relieved to see that repeated tests with a moisture meter showed the wall gradually drying to the same moisture level as areas that had not been wet.

The Selling Game

The NY/Enterprise CityHome HDfC claimed that the houses were not selling well, and that the energy-efficient features of the houses weren’t helping. However, the separate organization

assigned the task of marketing the houses had For Sale signs in the windows with the wrong phone number, and prospective purchasers had not been told that the houses were energy-efficient.

At some time after we adjusted the control settings for the boilers, someone went around to each of the buildings, took the covers off the controls to gain access to the adjustments underneath, and set them to overheat the houses, for reasons that are still unclear to us. The security guards staying in the houses until they were sold just opened the windows. After two cold months, the next utility bill arrived and gave everyone involved an excuse to say the buildings cost more to heat than a normal building.

All the houses were eventually bought by people whom we later visited to answer any questions that they might have. We explained the apparent paradox of spending money to tighten up the houses and then spending money to ventilate them by describing the inefficiency of “natural” ventilation. Our explanations, and the fact that the fan electricity cost only \$46.20 per year, convinced the owners to keep their ventilation fans running. All the owners understood the argument that \$46.20 is equivalent to only a few asthma inhalers.

The owners were generally enthusiastic to find out that they could control



Benedict used a blower door to test the air tightening in each apartment. In the finished units, the ACH at 50 Pa ranged from 2.3 to 6.7.

individual room temperatures. Several owners noted that the houses were unusually quiet. This is not surprising, as air sealing is considered to be an essential part of soundproofing. As the houses were built to exceed code minimum construction standards, they were built without air conditioning, but separate circuit breakers and outlets were provided for window air conditioners to comply with code. We encouraged owners not to buy air conditioners unless they actually felt uncomfortable in a room, explaining that the insulation and low-e windows would reduce the cooling load.

Since people are used to having a single thermostat on a wall, it was particularly difficult to explain the outdoor reset controls on the heating systems. Although some owners found these controls intimidating, any simpler alternative would have required multiple relays and more wiring, while significantly compromising on performance.

The Final Irony

Both New York City's Department of

Housing Preservation and Development and the lender, Community Preservation Corporation, specifically forbade Chris Benedict to design anything energy-efficient in next year's round of the CityHome project. They explained that they wanted to closely monitor energy use in these buildings for a few years before they decide whether to go energy-efficient again. To date, no monitoring has been done. NY/Enterprise CityHome does not seem interested in monitoring the buildings themselves. We would really like to see an independent party monitor these buildings and compare their energy use to that of rehabbed buildings that are not built to be energy-efficient. Ironically, the contractor expressed great relief at not being told to install unusual heating systems in the next round of CityHomes, but jumped at the chance to install direct-vent boilers to avoid the cost of chimneys and boiler rooms.

Numerous people conclude conversations about these buildings with talk of doing a "demonstration project."

We do not consider these houses to be a demonstration project. Why do you have to demonstrate what we and other people have already done?

We paid for the design and post-construction blower door testing of these houses because we think they're an example of what can be achieved when people do a good job and take responsibility for how their work affects the earth. In a twist that must rank very highly on the "irony of ironies" list, we may never get to do this type of work again for a city-sponsored housing project because our "demonstration project" was not made credible by government or not-for-profit funding. 

Henry Gifford is a boiler mechanic who lives in New York City.

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